

BRAIN TUMOR CLASSIFICATION USING HYBRID INTELLIGENCE

A Project Report submitted in partial fulfillment of the requirements for the award of the
degree of

**BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING
(Specialization in Artificial Intelligence & Machine Learning)**

Submitted by

**AKSHAT VASHISTH (VU22CSEN300234)
MANIKONDA SATYA VARDHAN (VU22CSEN300205)
MUPPASANI SHRUTHIKKA (VU22CSEN300300)**

Under the esteemed guidance of
Dr. G Venkata Sivanarayana
Assistant Professor – GITAM University



**Department of Computer Science and Engineering/ Department of
Artificial Intelligence and Data Science
GITAM School of CSE, GITAM (Deemed to be University),
Visakhapatnam.
2025**



DECLARATION

I hereby declare that the project report entitled **“BRAIN TUMOR CLASSIFICATION USING HYBRID INTELLIGENCE”** is an original work done in the Department of Artificial Intelligence and Data Science, GITAM School of CSE, GITAM (Deemed to be University) submitted in partial fulfillment of the requirements for the award of the degree of B.Tech. in Computer Science and Engineering (AI&ML). The work has not been submitted to any other college or University for the award of any degree or diploma.

Date: 28-11-2025

Registration Numbers	Names	Signature
VU22CSEN0300234	AKSHAT VASHISTH	
VU22CSEN0300205	MANIKONDA SATYA VARDHAN	
VU22CSEN0300300	MUPPASANI SHRUTHIKKA	

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING/
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA
SCIENCE
GITAM School of CSE
GITAM (Deemed to be University)**



CERTIFICATE

This is to certify that the project report entitled “**BRAIN TUMOR CLASSIFICATION USING HYBRID INTELLIGENCE**” is a bonafide record of work carried out by AKSHAT VASHISTH (VU22CSEN0300234), MANIKONDA SATYA VARDHAN (VU22CSEN0300205), MUPPASANI SHRUTHIKKA (VU22CSEN0300300) students submitted in partial fulfillment of requirement for the award of degree of Bachelors of Technology in Computer Science and Engineering (AI&ML).

Date: 27-11-2025

Project Guide

Dr. G Venkata Sivanarayana
Assistant Professor

Head of the Department

Dr. K Naveen Kumar
Associate Professor, HOD
Department of Artificial Intelligence
and Data Science

ACKNOWLEDGEMENT

We express our sincere gratitude to Dr. G Venkata Sivanarayana, Assistant Professor, Department of Computer Science Engineering, for his valuable guidance, continuous support, and encouragement throughout the development of our capstone project titled “Brain Tumor Classification using Hybrid Intelligence”. His insightful suggestions, motivation, and constant supervision greatly contributed to the successful completion of this work.

We extend our heartfelt thanks to our reviewers, Dr. P N R L Chandra Sekhar, Associate Professor, Department of Computer Science Engineering, and Dr. Gurpreet Singh Chhabra, Assistant Professor, Department of Computer Science Engineering, for their constructive feedback, evaluation, and valuable recommendations during the review process. Their inputs helped us refine and strengthen our project.

We are grateful to the Department of Artificial Intelligence and Data Science, GITAM School of CSE, GITAM (Deemed to be University), for providing the resources, academic environment, and facilities required for carrying out this project. We also thank the faculty members and friends who have supported us throughout this work.

TABLE OF CONTENTS

S.No.	Description	Page No.
1.	Abstract	1
2.	Introduction	2
3.	Literature Review	3
4.	Problem Identification & Objectives	6
5.	Existing System	7
6.	Proposed System	9
7.	System Architecture	10
8.	Tools/Technologies Used	12
9.	Conclusion	13
10.	References	14

ABSTRACT

Accurate detection and classification of brain tumors from MRI images remain challenging due to variations in tumor size, location, and intensity. Deep learning models such as Convolutional Neural Networks (CNNs) have shown strong capability in automated tumor analysis; however, their performance is highly dependent on optimal hyperparameter selection and network architecture design. Traditional optimization methods such as Particle Swarm Optimization (PSO) face significant limitations, including restricted search spaces due to fixed initialization ranges and the risk of premature convergence toward local optima. These drawbacks prevent PSO from fully exploring the architectural and training parameter space, thereby limiting segmentation accuracy. To address these challenges, this work proposes a hybrid optimization framework that integrates PSO with a Genetic Algorithm (GA) for enhanced brain tumor segmentation. The pipeline begins with MRI preprocessing followed by feature extraction using a CNN-based encoder. PSO is then employed to optimize continuous hyperparameters such as learning rate, dropout rate, filter sizes, and regularization coefficients. While PSO provides effective fine-tuning within a confined numeric space, GA is subsequently utilized to evolve discrete and structural components of the network, including the number of convolutional layers, activation functions, pooling strategies, and architectural connectivity. By combining PSO's rapid continuous search with GA's population-based structural exploration, the hybrid model overcomes the individual limitations of each technique and achieves a more global, flexible, and robust optimization.

INTRODUCTION

Medical imaging plays a critical role in the early detection and diagnosis of brain tumors, supporting timely treatment planning and improving patient survival. Among available modalities, Magnetic Resonance Imaging (MRI) is preferred due to its high contrast resolution and detailed anatomical representation. However, manually analyzing MRI scans is time-consuming, subjective, and prone to error, especially when tumor margins are unclear or vary in shape, size, and appearance. These challenges highlight the need for reliable, automated brain tumor segmentation and classification systems.

Early computational methods used handcrafted image processing techniques such as thresholding, GLCM, LBP, clustering, and edge detection. The emergence of deep learning, particularly Convolutional Neural Networks (CNNs), has significantly improved medical image analysis because CNNs automatically extract hierarchical visual features. Despite this, CNN performance depends heavily on appropriate selection of hyperparameters and architecture design, including learning rate, number of layers, filter sizes, activation functions, and dropout rate. Improper configurations can cause slow convergence, overfitting, or reduced generalization, making manual tuning unreliable and inefficient.

To improve tuning, optimization algorithms such as Particle Swarm Optimization (PSO) have been explored. PSO can automatically adjust continuous CNN hyperparameters; however, it is limited by predefined parameter ranges and prone to premature convergence, often failing to explore more complex architectural combinations. Similarly, Genetic Algorithms (GA) are useful for evolving network structures but are less effective at fine-tuning continuous parameters. These limitations create a research gap and motivate the need for a hybrid strategy.

The proposed work, titled “**Brain Tumor Segmentation Using Hybrid Intelligence (PSO + GA)**”, integrates both optimization methods to overcome their individual drawbacks. In this framework, PSO is used to optimize continuous hyperparameters, while GA modifies discrete architectural components such as number of layers, pooling strategies, and activation functions. This combination enables broader search capability, improved model generalization, and better convergence toward an optimal CNN architecture.

LITERATURE REVIEW

Naqvi *et al.* [1] proposed a CNN architecture optimization framework using Particle Swarm Optimization (PSO) for automated brain tumor classification. In their approach, PSO was used to tune hyperparameters such as activation functions, number of convolutional filters, and kernel size to enhance classification performance. The PSO-driven CNN showed improved accuracy compared to manually tuned architectures, demonstrating the potential of optimization-driven model development. However, the system relied on predefined parameter ranges, limiting scalability, and the algorithm frequently converged prematurely to local optima. Furthermore, the architecture itself remained fixed during optimization, restricting structural flexibility and preventing deeper or alternative CNN designs.

Deepthi and Jyothi [2] explored classical computer vision methods for brain tumor detection, using handcrafted texture descriptors such as GLCM, LBP, and HoG, combined with supervised classifiers like Support Vector Machines and Random Forests. Their system performed reasonably well on controlled datasets and small image samples. However, handcrafted feature extraction lacked robustness to noise, brightness changes, and shape variations in tumors. Additionally, the CNN models used in their study were shallow and manually configured without any optimization strategy, resulting in inconsistent performance and limited generalizability across datasets.

Xie and Yuille [3] introduced a Genetic Algorithm (GA)-based evolutionary method for CNN architecture search. Their work demonstrated that GA could evolve deep learning models by applying selection, crossover, and mutation, successfully building architectures that outperform manually designed CNNs. The approach showed strong adaptability and structural innovation, enabling automated design of non-standard neural architectures. However, GA struggled to optimize continuous hyperparameters such as learning rate and dropout rate and required high computational time due to repeated model training. The method's performance also depended heavily on the diversity and quality of the initial population.

Molina *et al.* [4] investigated MRI-based brain tumor segmentation using encoder-decoder architectures such as U-Net and V-Net, leveraging skip connections and deep convolutional blocks to preserve spatial context. Their models achieved strong segmentation quality and demonstrated reliability in identifying tumor boundaries. However, the architecture required extensive manual

hyperparameter tuning and dataset-specific adjustments, limiting reproducibility and automation. Additionally, the fixed architecture design lacked adaptability and performance declined when applied to datasets with different imaging properties.

Shanthi and Sabeenian [5] proposed a threshold-based segmentation and classification system that relied on intensity histogram analysis and fuzzy logic rules. Their method offered simplicity and low computational cost, making it suitable for smaller medical datasets. However, due to its sensitivity to pixel intensity variations and anatomical noise, segmentation accuracy was highly inconsistent, especially when tumors had diffused boundaries or low contrast in MRI scans.

Ismael and Hefny [6] introduced a transfer-learning-based approach using fine-tuned pre-trained CNN architectures such as ResNet and VGG for brain tumor detection. Their method reduced model training time and showed improved accuracy compared to classical machine learning systems. Despite its effectiveness, the approach required manual hyperparameter adjustment and lacked optimization strategies for adapting the model architecture to medical-specific patterns, making it sensitive to overfitting on limited Medical Imaging datasets.

Saxena *et al.* [7] performed a comparative study of multiple deep CNN models for tumor segmentation and classification, evaluating performance across dataset size, depth, and architecture complexity. Their findings indicated that deeper models achieved higher accuracy but also required significantly more computational power and careful layer-wise tuning. The absence of automated architecture search resulted in trial-and-error-based experimentation, making the process time-consuming and inefficient.

Alfonse and Salem [8] implemented a Random Forest-based classifier using GLCM feature vectors extracted from MRI scans. Their system achieved satisfactory accuracy and required fewer training samples compared to deep learning models. However, the reliance on handcrafted features limited adaptability and performance degraded significantly for complex or irregular tumor shapes, demonstrating the need for automated feature learning.

Real *et al.* [9] demonstrated a large-scale evolutionary architecture search technique capable of designing high-performance CNN models without human intervention. Their approach showed that evolutionary methods can outperform conventional architecture design in deep learning. However,

computational cost was extremely high, requiring multiple GPUs and extensive training cycles, limiting scalability for smaller research environments.

Agarwal and Tripathi [10] proposed a hybrid deep learning framework combining CNN feature extraction with Support Vector Machines (SVM) for classification of brain tumor MRI scans. Their approach demonstrated improved accuracy over standalone CNN and traditional machine learning models, particularly in distinguishing tumor subtypes. However, the system still relied on manually selected hyperparameters and did not include any automated architecture search or evolutionary optimization. As a result, model tuning required iterative experimentation, making the process time-consuming and less scalable.

Lakshmanan *et al.* [11] developed a PSO-enhanced transfer learning approach where pretrained deep CNN architectures such as AlexNet and Inception were fine-tuned using PSO-based hyperparameter search. The hybrid system showed improved classification accuracy and faster convergence compared to manual tuning strategies. Despite promising results, the study was limited to fine-tuning existing architectures rather than generating new ones. Additionally, the PSO algorithm faced local optima issues, and the system did not address discrete architecture selection, limiting model adaptability for different medical imaging scenarios.

Khan *et al.* [12] introduced a hybrid optimization approach combining Differential Evolution (DE) with Convolutional Neural Networks to classify brain tumors from MRI images. Their system automatically fine-tuned hyperparameters such as learning rate, batch size, and filter count, demonstrating improved performance compared to manually tuned and single-optimizer deep learning models. However, the method focused primarily on numerical hyperparameters and did not incorporate structural or architectural optimization. As a result, network depth, activation layers, and pooling strategies remained fixed, limiting adaptability and preventing exploration of potentially more efficient architectures for diverse medical imaging datasets.

PROBLEM IDENTIFICATION & OBJECTIVE

Problem Identification

Brain tumor segmentation from MRI images is a complex and challenging task due to the irregular structure, varying intensity levels, noise, and overlapping boundaries between tumor and normal tissue. Existing systems either rely on traditional image processing methods, which fail to capture deep spatial features, or deep learning methods that require carefully tuned hyperparameters and well-designed architectures to perform effectively. Manual tuning of CNN hyperparameters is time-consuming, inconsistent, and depends heavily on expertise. Optimization techniques such as PSO can improve parameter selection, but they are limited by fixed parameter ranges and tend to converge prematurely, leading to suboptimal results. Similarly, GA can evolve architecture but struggles with continuous parameter tuning. These limitations highlight the need for a hybrid, automated, and intelligent optimization framework capable of improving both CNN hyperparameters and architecture to enhance tumor segmentation accuracy and model robustness.

Objectives of the Project

- To automate brain tumor detection and segmentation from MRI images by applying deep learning with preprocessing and CNN-based feature extraction.
- To optimize CNN performance by applying Particle Swarm Optimization (PSO) for continuous hyperparameters and Genetic Algorithm (GA) for architecture-level parameters.
- To integrate PSO and GA into a hybrid optimization framework for improved model accuracy, robustness, and global search capability.
- To evaluate and compare the optimized model using performance metrics such as accuracy, Dice score, and IoU against standard CNN and PSO-only models.

EXISTING SYSTEM

Brain tumor classification and segmentation have traditionally relied on deep learning models such as Convolutional Neural Networks (CNNs), which automatically extract hierarchical features from MRI images. The existing system described in the research paper you referred to uses Particle Swarm Optimization (PSO) to automatically design an optimal CNN architecture for classifying brain tumors into different types using MRI images. In this system, each particle in the swarm represents one possible CNN model configuration—defined by parameters such as number of convolutional layers, filter size, number of filters, pooling layers, and fully connected layers. The PSO algorithm evaluates each architecture by partially training it on the dataset and computing accuracy or loss as a fitness measure. The particle that achieves the highest accuracy becomes the global best (gBest), and the swarm iteratively moves toward this best-performing configuration. This allows the algorithm to automatically discover CNN architectures without manual tuning or deep learning expertise.

The existing system successfully demonstrates that PSO can automate CNN design and achieve high accuracy (up to 99.19% according to the paper). The architecture search process is guided by PSO's exploration (random movement) and exploitation (moving toward pBest and gBest). The method reduces manual effort and enables CNN models to be generated automatically based on dataset-specific patterns. The PSO-based CNN architecture generation is particularly helpful for domains such as medical image analysis where small variations in hyperparameters can significantly influence diagnostic accuracy. The study further proves that evolutionary algorithms can outperform manually designed CNNs for MRI brain tumor classification.

However, despite these advantages, the existing system has multiple limitations and constraints, many highlighted directly in the paper, which reduce its flexibility and generalization capability in real-world scenarios.

Drawbacks of the Existing System

The first major limitation of the PSO-based CNN optimization framework is that hyperparameter tuning is confined to fixed search boundaries defined during initialization. Since particle exploration is limited by predefined upper and lower bounds for parameters such as filter size, kernel

dimensions, and number of layers, the algorithm cannot discover architectures beyond the manually selected ranges, restricting flexibility and preventing the discovery of potentially superior configurations.

A second drawback is the tendency of the PSO algorithm to prematurely converge toward local optima. As reported in the study, PSO rapidly collapses toward early high-performing particles, reducing population diversity and exploration capability. This stagnation prevents the algorithm from effectively searching the broader architectural space, especially when tuning deep or complex CNN structures, making the optimization process highly dependent on initial particle configurations.

A third key limitation is that PSO is suitable for optimizing continuous-valued parameters but is inefficient in handling discrete architectural decisions. The system cannot automatically explore design variations such as activation function choice, pooling strategy, or layer addition and removal. As a result, architecture design relies on static rules rather than dynamic search, leading to limited scalability, restricted model diversity, and suboptimal network evolution for complex datasets.

PROPOSED SYSTEM

The proposed system introduces a Hybrid Intelligence Framework that integrates Particle Swarm Optimization (PSO) and Genetic Algorithm (GA) to design and optimize a Convolutional Neural Network (CNN) for brain tumor segmentation/classification from MRI images.

The system begins with preprocessing steps such as skull stripping, noise removal, normalization, and contrast enhancement to improve MRI image quality. A CNN-based feature extractor then generates multi-level feature maps that capture edges, intensity variations, and potential tumor regions. These features serve as the input to the hybrid optimization framework.

The first stage of optimization uses PSO, which fine-tunes continuous hyperparameters such as learning rate, dropout rate, kernel size, number of filters, and regularization values. Each particle represents one possible hyperparameter configuration, and its fitness is computed by training a CNN for a few epochs on the feature maps. Through velocity and position updates, PSO identifies the best continuous parameter set (PSO_best). However, unlike the existing system, the proposed framework does not stop at PSO because PSO alone cannot efficiently explore network architecture or avoid local optima.

To overcome these limitations, the system incorporates a second optimization stage using a Genetic Algorithm, which evolves discrete structural components of the CNN, such as number of convolutional layers, activation functions, pooling strategy, and structural variations. GA uses PSO_best as a starting point and generates multiple architecture variations using selection, crossover, and mutation. Each candidate architecture is evaluated by training using the PSO-optimized hyperparameters, and the best architecture (GA_best) is selected after several generations. This dual-stage optimization ensures that the final CNN is optimized both horizontally (hyperparameters) and vertically (architecture).

Finally, the system constructs a CNN model using the combined best hyperparameters and architecture obtained from PSO and GA. This optimized CNN is fully trained on the MRI dataset using backpropagation and then tested on unseen data to generate tumor segmentation masks or classification outputs.

SYSTEM ARCHITECTURE

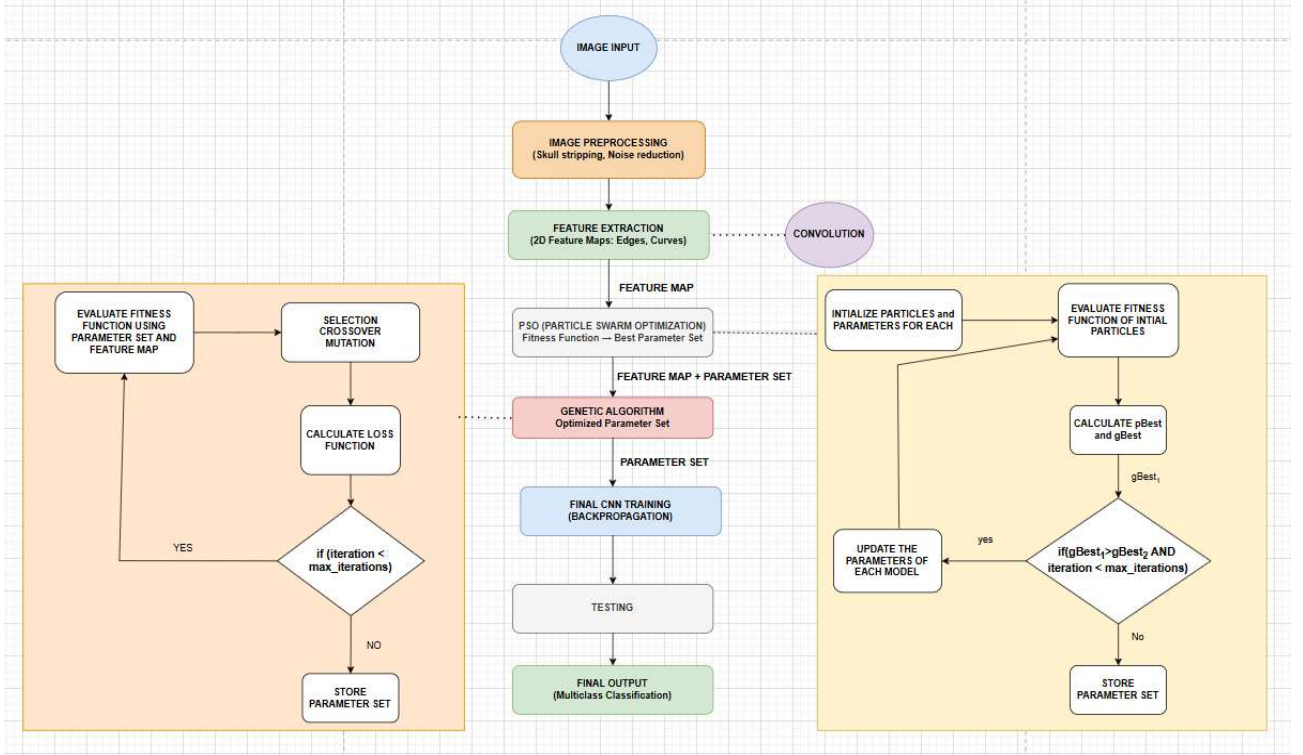


Figure 1: System Architecture of the Hybrid Intelligence Model

The proposed system architecture is designed to automate and optimize brain tumor segmentation and classification from MRI images using a hybrid PSO–GA optimized Convolutional Neural Network (CNN). The architecture is composed of sequential processing modules, each performing a specific function in the workflow. The system begins with the Image Input Module, where MRI scans from datasets such as BRATS are loaded. These images may vary in size, contrast, and orientation, and therefore must be standardized before analysis.

Following image acquisition, the data is passed into the Preprocessing and Feature Extraction module. This stage applies essential preprocessing procedures, including skull stripping, noise reduction, intensity normalization, contrast enhancement, and resizing, ensuring consistent image quality across the dataset. Once preprocessing is complete, a CNN-based feature extractor automatically analyzes the image and produces multi-channel feature maps. These feature maps highlight relevant structural patterns such as intensity changes, textures, edges, and potential tumor regions, forming the input for subsequent optimization processes.

The first optimization phase is conducted using Particle Swarm Optimization (PSO), which focuses on tuning continuous hyperparameters of the CNN model. During this phase, PSO initializes a population of particles, where each particle represents a unique hyperparameter configuration. Each configuration is used to build and train a CNN for a limited number of epochs, and its performance is evaluated using metrics such as accuracy or Dice similarity score. Based on the fitness results, PSO updates the velocity and position of particles using personal best (pBest) and global best (gBest) feedback. Iterations continue until convergence criteria are met, resulting in a set of optimal continuous hyperparameters (PSO_best), which is then forwarded to the next optimization stage.

The Genetic Algorithm (GA) module represents the second optimization stage and is responsible for evolving the discrete architectural components of the CNN. Unlike PSO, GA modifies elements such as the number of convolutional and dense layers, activation functions, pooling techniques, and filter expansion patterns. The process begins with an initial population of architectural variations. Each candidate is evaluated using the hyperparameters determined in the PSO stage. GA then performs selection, crossover, and mutation to refine architecture quality over multiple generations. This iterative evolution process concludes when an optimized neural architecture (GA_best) is identified.

Once both optimization stages are completed, the Final CNN Construction module combines the best hyperparameters identified by PSO with the optimal architecture produced by GA. This results in a highly customized CNN design tailored specifically for brain tumor analysis. The constructed model is then trained fully using backpropagation, where forward propagation generates predictions, loss functions compute error, and gradients fine-tune internal model weights. This training results in a fully optimized deep learning model capable of learning and recognizing complex tumor characteristics.

After training, the model undergoes evaluation in the Testing Module, where performance is assessed using unseen MRI scans. Standard medical imaging evaluation metrics, such as accuracy, Dice Similarity Coefficient (DSC), Intersection over Union (IoU), sensitivity, precision, and recall are calculated to verify model reliability and effectiveness. Finally, depending on the system's design objective, the output module generates either a segmentation mask that highlights tumor regions or a classification label identifying the tumor type as glioma, meningioma, pituitary tumor etc. These outputs can then be compared with ground truth labels or visualized for interpretation.

TOOLS & TECHNOLOGIES USED

- **Google Colab / Jupyter Notebook**
Google Colab and Jupyter Notebook served as the development and execution environments for training and testing the proposed hybrid model.
- **Programming Language: Python**
Python is the primary programming language used because of its strong ecosystem supporting machine learning, deep learning, and scientific computation.
- **Frontend Technologies: HTML, CSS, and JavaScript**
HTML, CSS, and JavaScript are used to develop the user interface for displaying predictions and segmentation results.
- **Backend Framework: Flask / Streamlit**
Flask or Streamlit is used to deploy the trained model and serve it through a lightweight web application. Flask supports API-based communication between the user interface and the model, while Streamlit simplifies direct machine learning model deployment with minimal development overhead.
- **Machine Learning & Deep Learning Libraries: Scikit-learn and TensorFlow**
Scikit-learn is used for preprocessing and TensorFlow provides the essential framework to build, train, and optimize the CNN architecture used for brain tumor classification.
- **Data Handling Libraries: Pandas and NumPy**
Pandas and NumPy are used for managing MRI metadata, processing numerical arrays, and performing matrix operations essential for neural network computations.
- **Visualization Tools: Matplotlib and Seaborn**
Matplotlib and Seaborn are used to visualize model training behavior, including accuracy curves, loss graphs, segmentation overlays, and PSO/GA optimization results.
- **GitHub**
Provides version control, collaborative development, and secure storage of project files. It ensures proper tracking of updates and allows seamless teamwork during implementation.

CONCLUSION

The proposed hybrid PSO–GA optimized CNN framework is designed to address key limitations identified in existing PSO-based brain tumor detection systems, including restricted parameter search ranges, premature convergence, and limited capability to optimize discrete architectural variations. By separating the optimization process into two complementary stages—PSO for continuous hyperparameter tuning and GA for architecture-level refinement—the methodology enables a broader and more systematic exploration of possible model configurations. This conceptual framework lays the foundation for building a more adaptable and scalable deep learning-based system for automated brain tumor segmentation and classification, and it also provides a structured direction for further development, implementation, and performance validation in future stages of the project.

REFERENCE

- [1] R. A. Naqvi, A. U. Haq, M. Z. Khan, M. S. Farid, and M. K. Shahzad, “Automatic CNN architecture optimization for brain tumor classification using Particle Swarm Optimization,” *Journal of Imaging*, vol. 11, no. 31, pp. 1–19, 2023.
- [2] M. R. Deepthi and M. Jyothi, “Brain Tumor Detection in MRI Images Using Texture Features and SVM,” *International Journal of Engineering Research & Technology*, vol. 9, no. 6, pp. 450–455, 2020.
- [3] A. S. Alfonse and M. A. Salem, “An Automatic Classification of Brain Tumors from MRI Using GLCM and Random Forest,” *International Journal of Computer Applications*, vol. 146, no. 9, pp. 1–8, 2016.
- [4] Y. Xie and A. Yuille, “Genetic CNN,” in *Proc. IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 1379–1388.
- [5] E. Real, S. Moore, A. Selle *et al.*, “Large-Scale Evolution of Image Classifiers,” in *Proc. 34th International Conference on Machine Learning (ICML)*, 2017, pp. 2902–2911.
- [6] J. Suganuma, S. Shirakawa, and T. Nagao, “A Genetic Programming Approach to Designing CNN Architectures,” in *GECCO '17: Genetic and Evolutionary Computation Conference*, 2017, pp. 497–504.
- [7] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional Networks for Biomedical Image Segmentation,” in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015, pp. 234–241.
- [8] R. Ranjbarzadeh, A. B. Kargari, S. J. Ghouschi, S. Anari, M. Naseri *et al.*, “Brain tumor segmentation based on deep learning and an attention mechanism using MRI multi-modalities,” *Sci. Rep.*, vol. 11, Article 10930, 2021.
- [9] A. Mahmoud, D. Ghazi, M. Khan *et al.*, “Advanced deep learning approaches for accurate brain tumour classification on MRI images,” *Symmetry*, vol. 15, no. 3, art. 571, 2023

- [10] P.K. Ramtekkar, “Accurate detection of brain tumour using optimized feature and CNN-based classifier,” *Multimedia Tools and Applications*, 2023.
- [11] S. Akkus, A. Galimzianova, A. Hoogi, D. L. Rubin, and B. Erickson, “Deep learning for brain MRI segmentation: State of the art and future directions,” *J. Digit. Imag.*, vol. 30, no. 4, pp. 449–459, 2017.
- [12] S. Bauer, R. Wiest, L. P. Nolte, and M. Reyes, “A survey of MRI-based medical image analysis for brain tumor studies,” *Phys. Med. Biol.*, vol. 58, no. 13, pp. 97–129, 2013.
- [13] F. Isensee, P. Kickingereder, W. Wick, M. Bendszus, and K. H. Maier-Hein, “No new-net: U-Net architecture with minor improvements provides state-of-the-art results for brain tumor segmentation,” in *BrainLes Workshop, MICCAI*, 2018, pp. 234–244.
- [14] A. Perez-Borrero, D. F. Silva, J. A. Ramos, and A. Conci, “A review of deep learning methods for brain tumor segmentation,” *Appl. Sci.*, vol. 10, no. 6, art. 1993, 2020.